

3 Problem formulation

This study formulates wave composition as a finite selection problem coupled with a fixed operational evaluator. A decision instance contains an order pool, a fixed wave size, a warehouse configuration, and a specified family of elevator models. The tactical decision selects either a concrete candidate wave or a structural wave class. A canonical serial resource-reservation rule then maps the selected orders to AMR and elevator completion times. The resulting formulation separates the composition decision from the operational mechanism used to evaluate it.

3.1 Indices and sets

The basic indices and their domains are listed below. Candidate-wave construction and structural-class membership are defined in Sections 3.2.2 and 3.2.3, respectively.

Symbol	Definition
$f \in \mathcal{F} = \{1, \dots, F\}$	warehouse-floor index
$o \in \mathcal{O}$	order in the available order pool
$a \in \mathcal{A} = \{1, \dots, A\}$	AMR index
$e \in \mathcal{E} = \{1, \dots, E\}$	physical-elevator index under M_2 and M_3
$v \in \mathcal{V}_1 = \{1, \dots, Ec\}$	virtual single-request elevator-slot index under M_1
$k \in \mathcal{K} = \{1, \dots, K\}$	candidate-wave index, common to all evaluators
$j \in \mathcal{J}_n = \{1, \dots, n\}$	position index in the canonical processing sequence
$q \in \mathcal{Q}$	selectable structural-class label
$m \in \mathcal{M} = \{M_1, M_2, M_3\}$	elevator-evaluator index

3.2 Parameters

3.2.1 Warehouse, order, and service parameters

Symbol	Definition
F	number of warehouse floors
A	number of AMRs
E	number of physical elevator cars
c	maximum AMR occupancy of one physical elevator trip
n	fixed number of orders in a wave
K	number of generated candidate waves
t_w	common decision and release epoch of the selected wave
f^0	common initial floor of the AMRs and elevator resources
t_o	immutable and unique identifier of order o
s_o	source floor of order o
d_o	destination floor of order o
$r_o \geq 0$	order-ready offset relative to t_w
γ	elevator travel time per traversed floor
τ^L	elevator loading time
τ^U	elevator unloading time
τ^P	AMR pickup-service time
τ^D	AMR drop-off-service time
σ	lognormal phase-time noise parameter under M_3

α | lower-tail probability used to construct the structural classes

Order o becomes eligible at $t_w + r_o$. Thus, r_o is an observed order attribute at the wave decision epoch. The publication-scale evaluator uses $f^0 = 1$, $\gamma = 5$ s per floor, $\tau^L = \tau^U = 2$ s, and $\tau^P = \tau^D = 5$ s. The principal stochastic specification uses $\sigma = 0.20$.

3.2.2 Candidate waves and canonical sequence

The candidate pool contains K generated waves indexed by $\mathcal{K}$. For each $k \in \mathcal{K}$, let $W^k \subseteq \mathcal{O}$ denote the unordered set of orders in candidate wave k , with $|W^k| = n$. Its operational processing sequence is determined by:

$$\pi(W^k) = (o_1^k, \dots, o_n^k) = \text{sort}_{o \in W^k} (r_o, t_o) \quad (1)$$

where sorting is lexicographic: orders are first ranked by ready time offset r_o , and equal offsets are resolved by immutable order identifier t_o . All three elevator evaluators use this sequence.

For use in the candidate-selection constraints, the composition of W^k is represented by the derived incidence parameter

$$\delta_{ok} = \begin{cases} 1, & o \in W^k \\ 0, & o \notin W^k \end{cases} \quad (2)$$

3.2.3 Wave descriptors and structural classes

For wave W , define the source and destination endpoints located on floor f as:

$$p_f(W) = \frac{1}{2|W|} \sum_{o \in W} [\mathbb{I}(s_o = f) + \mathbb{I}(d_o = f)] \quad (3)$$

The endpoint-dispersion descriptor is:

$$C(W) = - \sum_{f \in \mathcal{F}} p_f(W) \log p_f(W) \quad (4)$$

Higher C indicates that order endpoints occupy a broader set of floor labels. Physical floor-to-floor distance enters the operational evaluator through $|s_o - d_o|$.

Let:

$$N^\uparrow(W) = \sum_{o \in W} \mathbb{I}(d_o > s_o), \quad N^\downarrow(W) = \sum_{o \in W} \mathbb{I}(d_o < s_o). \quad (5)$$

Directional imbalance is:

$$I(W) = \begin{cases} \frac{|N^\uparrow(W) - N^\downarrow(W)|}{N^\uparrow(W) + N^\downarrow(W)}, & N^\uparrow(W) + N^\downarrow(W) > 0, \\ [8pt] 0, & N^\uparrow(W) + N^\downarrow(W) = 0. \end{cases} \quad (6)$$

Same-floor orders do not enter either directional count. Let $\bar{r}_W = |W|^{-1} \sum_{o \in W} r_o$. Relative ready-time dispersion is:

$$T(W) = \begin{cases} \frac{\sqrt{|W|^{-1} \sum_{o \in W} (r_o - \bar{r}_W)^2}}{\bar{r}_W}, & \bar{r}_W > 0, \\ [10pt] 0, & \bar{r}_W = 0. \end{cases} \quad (7)$$

Lower T represents more compact ready offsets relative to their mean; higher T represents greater relative dispersion. The complete descriptor vector is:

$$\Phi(W) = (C(W), I(W), T(W)) \quad (8)$$

The principal corner design uses C and I . For a tail probability $\alpha \in (0, 0.5)$, let $Q_{u(X)}$ denote the empirical u -quantile of descriptor X in the candidate pool. Define

$$\begin{aligned} \mathcal{K}_{\text{HC}} &= \{k : C(W^k) \geq Q_{1-\alpha}(C)\}, \\ \mathcal{K}_{\text{LC}} &= \{k : C(W^k) \leq Q_{\alpha}(C)\}, \\ \mathcal{K}_{\text{HI}} &= \{k : I(W^k) \geq Q_{1-\alpha}(I)\}, \\ \mathcal{K}_{\text{LI}} &= \{k : I(W^k) \leq Q_{\alpha}(I)\} \end{aligned} \quad (9)$$

The four possible class labels are HH, HL, LH, and LL, where the first letter records the tail of C and the second records the tail of I . Their candidate-index supports are

$$\begin{aligned} \mathcal{K}_{\text{HH}} &= \mathcal{K}_{\text{HC}} \cap \mathcal{K}_{\text{HI}}, \\ \mathcal{K}_{\text{HL}} &= \mathcal{K}_{\text{HC}} \cap \mathcal{K}_{\text{LI}}, \\ \mathcal{K}_{\text{LH}} &= \mathcal{K}_{\text{LC}} \cap \mathcal{K}_{\text{HI}}, \\ \mathcal{K}_{\text{LL}} &= \mathcal{K}_{\text{LC}} \cap \mathcal{K}_{\text{LI}}. \end{aligned} \quad (10)$$

The structural-class label set is $\mathcal{Q} = \{q \in \{\text{HH}, \text{HL}, \text{LH}, \text{LL}\} : \mathcal{K}_q \neq \emptyset\}$. Quantile ties can place a candidate index in more than one support set, so the sets $\mathcal{K}_q$ may overlap. The main implementation uses $\alpha = 0.25$. Descriptor T is activated in the designated staggered-readiness analyses. For each $q \in \mathcal{Q}$, the fixed within-class candidate distribution is uniform on $\mathcal{K}_q$:

$$\nu_q(k) = \begin{cases} \frac{1}{|\mathcal{K}_q|}, & k \in \mathcal{K}_q, \\ 0, & k \notin \mathcal{K}_q. \end{cases} \quad (11)$$

After class q is selected, $\kappa \sim \nu_q$ denotes the candidate index drawn from $\mathcal{K}_q$, and W^κ is the wave released for execution.

3.3 Decision variables

Two formulations operate at different decision resolutions. The candidate formulation selects one concrete wave from the finite pool. The class formulation selects one structural class and applies the within-class rule in Equation (9).

Variable	Domain	Definition
z_k	$\{0, 1\}$	1 if candidate wave k is selected

x_o	$\{0, 1\}$	1 if order o belongs to the selected candidate wave
y_q	$\{0, 1\}$	1 if structural class q is selected
η^c	$\mathbb{R}_+$	largest candidate score over the specified evaluator family
η^o	$\mathbb{R}_+$	largest class-level median over the specified evaluator family

AMR assignments, elevator assignments, boarding decisions, and trip times are generated by the fixed evaluator. They are operational state quantities. The tactical optimization variables are z_k , x_o , and

y_q .

3.4 Assumptions

A1. Single-wave decision horizon. Each instance begins at t_w and ends when every order in the selected wave is complete. All performance measures refer to this isolated wave, and no other wave overlaps its evaluation interval.

A2. Known candidate information. The source floor, destination floor, ready offset, and immutable identifier of every available order are known when the candidate waves are generated.

A3. Canonical processing order. All evaluators process each candidate wave according to the common sequence $\pi(W^k)$ defined in Equation (1).

A4. Cold-start resource state. At t_w , every AMR and elevator resource is available at floor f^0 .

The instance begins without inherited resource reservations or unfinished orders.

A5. AMR mobility and service. An AMR remains assigned to an order through drop-off and rides the elevator during cross-floor movements. It may therefore serve orders on different floors. The principal evaluator aggregates intra-floor travel and handling into the fixed pickup and drop-off durations τ^p and τ^d . An elevator request is generated whenever the AMR's current floor differs from the next required floor.

A6. Fixed AMR assignment. Orders are considered in canonical order. Each order is assigned to the AMR with the smallest current availability time; an availability tie is resolved by AMR index. Order processing begins at the larger of the assigned AMR's availability time and $t_w + r_o$.

A7. Serial resource reservation. Each order is advanced through its complete route before the next canonical order is considered. Elevator requests update the selected resource state in this reservation order. The generated request timestamps need not be globally increasing.

A8. Stable elevator selection. Equal elevator availability is resolved by virtual-slot index under M_1 and by car index under M_2 and M_3 . When several cars have an open scheduled loading window for the same origin-destination request, the lowest-index boardable car is selected.

A9. Throughput abstraction M_1 . The E elevators with capacity c are represented by Ec independent single-request virtual slots. Each slot has its own location and availability clock. Separate requests occupy separate virtual trips.

A10. Co-occupancy evaluator M_2 . The evaluator contains E physical cars, each with capacity c . A request may join a scheduled trip when its origin and destination match, its request time is no later

than the trip's loading-end time, and residual capacity is available. A request that cannot join such a trip initiates a new reservation on the earliest-available car.

A11. Stochastic evaluator M_3 . Evaluator M_3 retains the assignment, boarding, and capacity rules of M_2 . The repositioning, loading, loaded-travel, and unloading durations of every new trip receive mean-one lognormal multipliers that are independent across phases and trips. For each phase multiplier Z ,

$$\log Z \sim \mathcal{N}\left(-\frac{\sigma^2}{2}, \sigma^2\right), \quad \mathbb{E}[Z] = 1. \quad (12)$$

The phase multipliers generated during one complete M_3 wave evaluation form a realization $\xi \in \Xi$, where Ξ is the corresponding realization space. Waiting time follows from the resource availability clock. Pickup and drop-off times retain their fixed values in the principal specification.

A12. Evaluator exogeneity. The resource configuration, operational rule, and evaluator parameters remain fixed while the tactical wave decision changes. Wave composition determines the evaluator request stream received by the evaluator.

A13. Fixed within-class implementation. Every selectable class contains at least one candidate. A selected class is converted to a concrete release by the distribution ν_q in Equation (9). The within-class candidate draw and the M_3 phase-time realization are independent.

3.5 Objective function

3.5.1 Wave makespan

Let $t_{jkm}^C(\xi)$ denote the completion time of the order in canonical position j of candidate k under evaluator m . The realized wave makespan is:

$$Y_{km}(\xi) = C_{\max}(W^k; m, \xi) = \max_{j \in \mathcal{J}_n} t_{jkm}^C(\xi) - t_W \quad (13)$$

The outcome is deterministic under M_1 and M_2 . Under M_3 , ξ indexes the phase-time realization.

3.5.2 Candidate-Wave benchmark

Let $\mathcal{M}_D = \{M_1, M_2\} \subset \mathcal{M}$ denote the deterministic evaluator family. For $m \in \mathcal{M}_D$, define the deterministic candidate score:

$$\ell_{km} = C_{\max}(W^k; m) \quad (14)$$

The single-evaluator finite-pool benchmark is

$$\min_{k \in \mathcal{K}} \ell_{km} \quad (15)$$

For a prespecified nonempty evaluator family $\emptyset \neq \mathcal{M}_R \subseteq \mathcal{M}_D$, the robust candidate benchmark is

$$\min_{k \in \mathcal{K}} \max_{m \in \mathcal{M}_R} \ell_{km} \quad (16)$$

The term pool optimum denotes the optimum obtained over the candidate waves indexed by $\mathcal{K}$.

3.5.3 Robust Class-Selection Objective

For class q and evaluator m , define

$$\mu_{qm}^{0.5} = \text{Med}_{\kappa \sim \nu_q, \xi} [Y_{\kappa m}(\xi)] \quad (17)$$

For M_1 and M_2 , the distribution in Equation (17) is induced by the within-class candidate rule. Under M_3 , it includes both candidate variation and stochastic trip durations. The superscript 0.5 identifies the median estimand. Equation (17) is the primary policy-level estimand. The median-minimax class problem over the deterministic evaluator family is

$$q^* \in \arg \min_{q \in \mathcal{Q}} \max_{m \in \mathcal{M}_D} \mu_{qm}^{0.5} \quad (18)$$

After q^* is selected, the release rule draws $\kappa \sim \nu_{q^*}$. Evaluator M_3 supplies the stochastic performance assessment of the selected class.